

Laboratory diagnosis of malaria

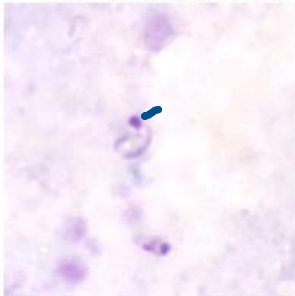
Plasmodium malariae**Basic guidelines**

- A. Capillary blood should be obtained by fingerstick, or venous blood should be obtained by venipuncture.
- B. Blood smears, at least two thick and two thin, should be prepared as soon as possible after collection. *Delay in preparation of the smears can result in changes in parasite morphology and staining characteristics.*
- C. Schüffner's dots can be demonstrated in Giemsa stain, which is preferred to Wright or Wright-Giemsa stains.

In *P. malariae* infections, red blood cells (rbcs) are normal or smaller than normal (3/4x) in size.

1. Rings

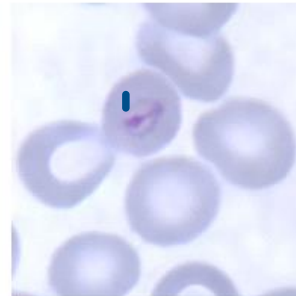
P. malariae rings have sturdy cytoplasm and a large chromatin dot.



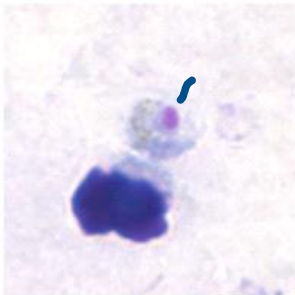
Ring in a thick blood smear.



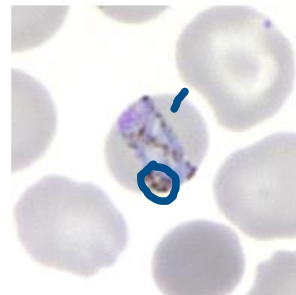
Rings in thin blood smears.

**2. Trophozoites**

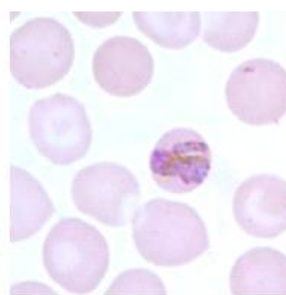
P. malariae trophozoites have compact cytoplasm and a large chromatin dot. Occasional band forms and/or "basket" forms with coarse, dark-brown pigment can be seen.



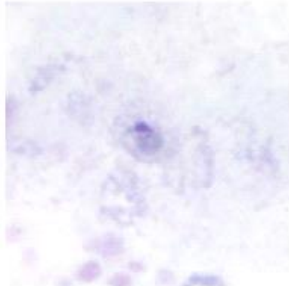
Trophozoite in a thick blood smear.



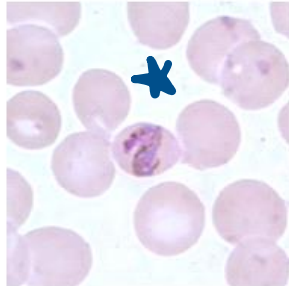
Band-form trophozoites in thin blood smears.



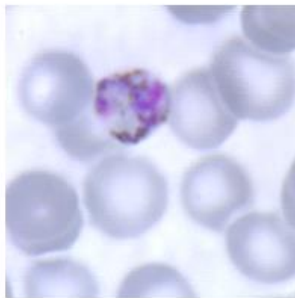
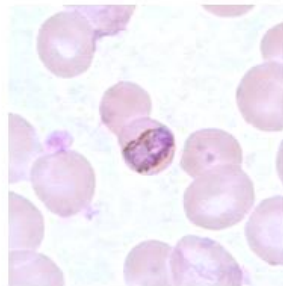
Plasmodium malariae



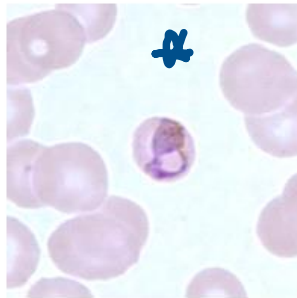
Trophozoite in a thick smear.



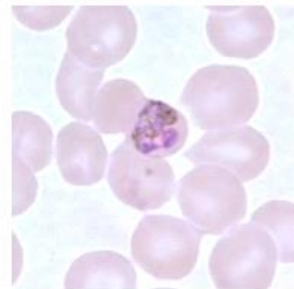
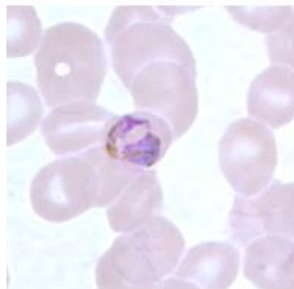
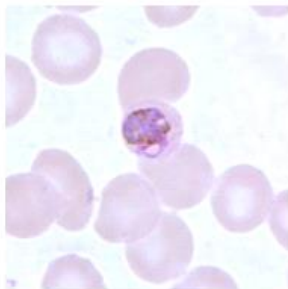
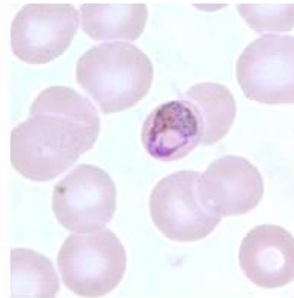
Band-form trophozoites in thin blood smears.



"Basket-form" trophozoite in a thin



"Basket-form" trophozoites in a thin smear.

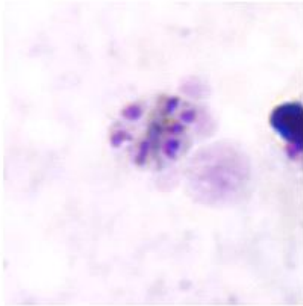


Trophozoites in thin blood smears. These images show variations on the "basket-form."

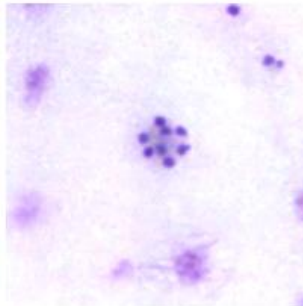
Plasmodium malariae

3. Schizonts

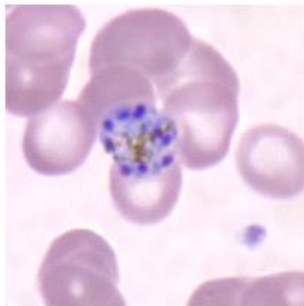
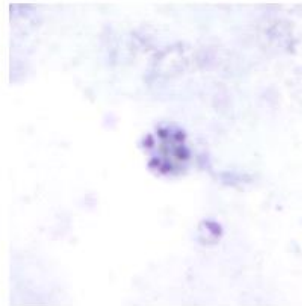
P. malariae schizonts have 6 to 12 merozoites with large nuclei, clustered around a mass of coarse, dark-brown pigment. Merozoites can occasionally be arranged as a rosette pattern.



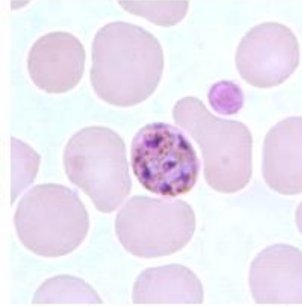
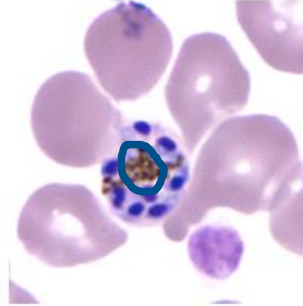
Schizont in a thick blood smear.



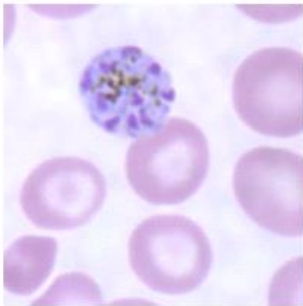
Schizont in thick blood smears. Note the classic "rosette" appearance of the merozoites.



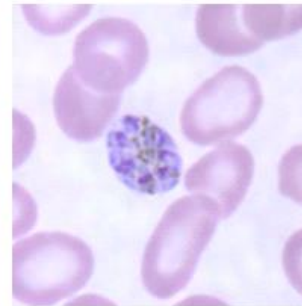
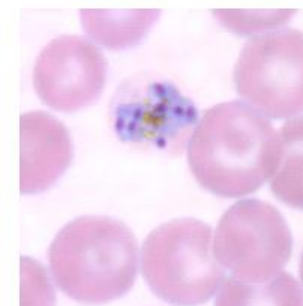
Schizonts in thin blood smears. The schizont on the left has the appearance of a rosette pattern.



Schizont in a thin blood smear; note the rosette pattern of the merozoites.



Schizonts in thin blood smears

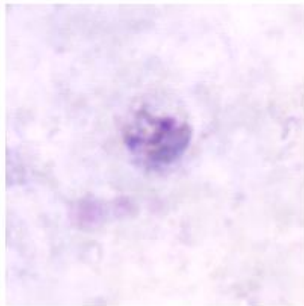


Schizont in a thin blood smear; another resembling a rosette.

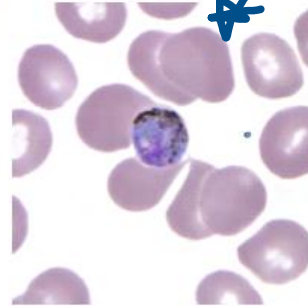
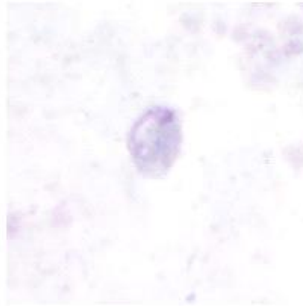
Plasmodium malariae

4. Gametocytes

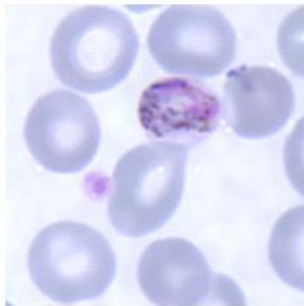
P. malariae gametocytes are round to oval with scattered brown pigment; they may almost fill the infected red blood cell.



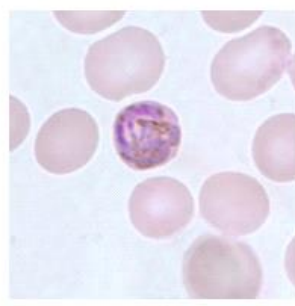
Gametocytes in thick blood smears.



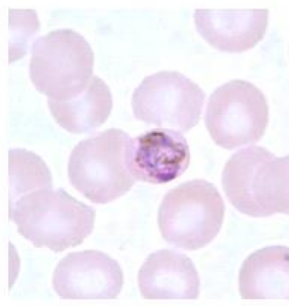
Gametocyte in a thin blood smear.



Gametocyte in thin blood smears.



Gametocytes in thin blood smears.



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Focus

Plasmodium ovale

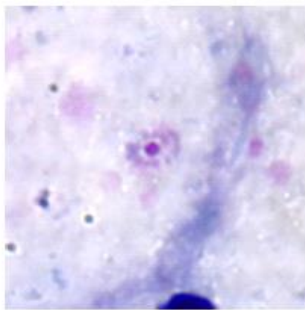
Basic guidelines

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- B. Blood smears, at least two thick and two thin, should be prepared as soon as possible after collection. *Delay in preparation of the smears can result in changes in parasite morphology and staining characteristics.*
- C. Schüffner's dots can be demonstrated in Giemsa stain, which is preferred to Wright or Wright-Giemsa stains.

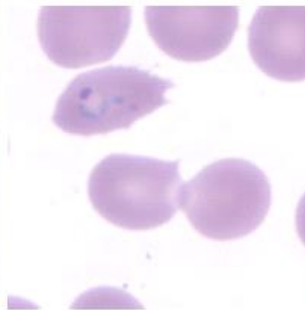
In *P. ovale* infections, red blood cells (rbcs) can be normal or slightly enlarged (up to 1 1/4x) in size, may be round to oval, and are sometimes fimbriated. Under optimal conditions, Schüffner's dots may be seen in Giemsa stained slides.

1. Rings

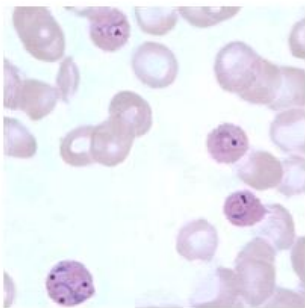
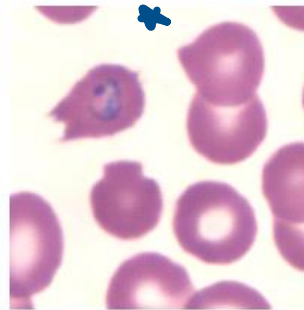
P. ovale rings have sturdy cytoplasm and large chromatin dots.



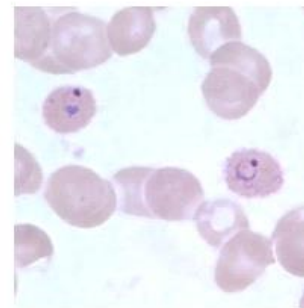
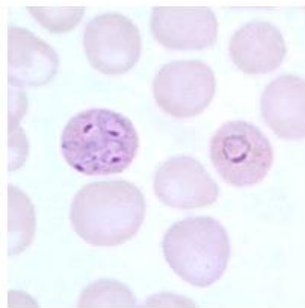
Ring in a thick blood smear.



Rings in fimbriated rbcs in thin blood smears.



Rings in thin blood smears. Note the multiply-infected rbcs.



Rings in a thin blood smear.

Plasmodium ovale

2. Trophozoites

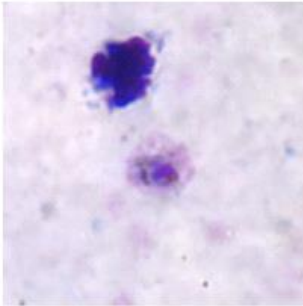
P. ovale trophozoites have sturdy cytoplasm, large chromatin dots, and can be compact to slightly irregular.



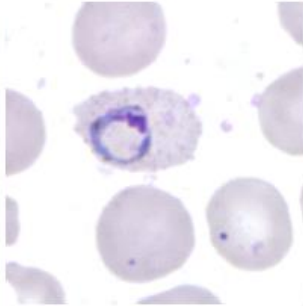
Plasmodium ovale

2. Trophozoites

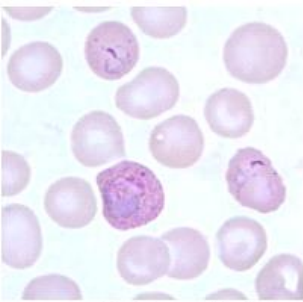
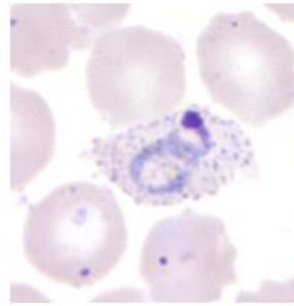
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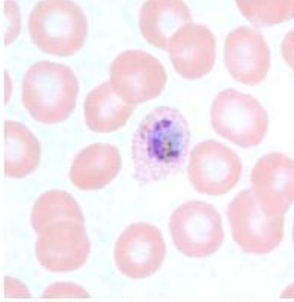
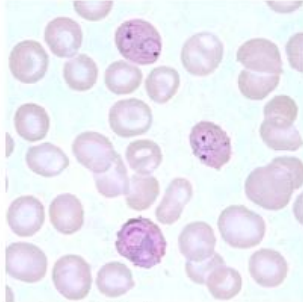
Trophozoite in a thick blood smear.



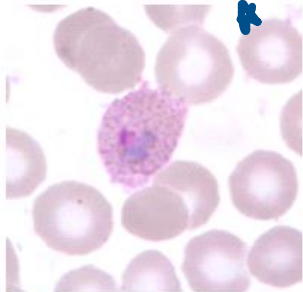
Compact trophozoites in fimbriated rbc's in thin blood smears. Schüffner's dots are also visible.



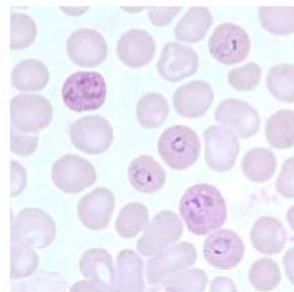
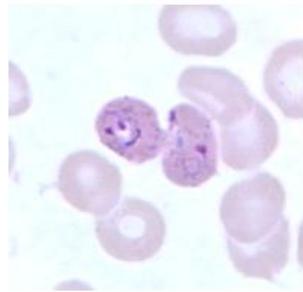
Ring forms and developing trophozoites in thin blood smears.



Compact trophozoite in a fimbriated rbc in a thin blood smear. Schüffner's dots are also visible.



Compact trophozoites showing Schüffner's dots. The image on the left also shows prominent fimbriation.



Ring forms, developing and compact trophozoites in a thin blood smear.

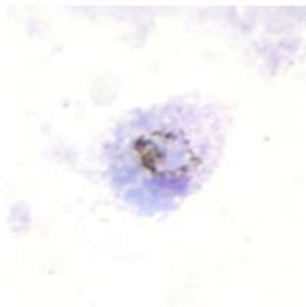
Plasmodium ovale

3. Gametocytes

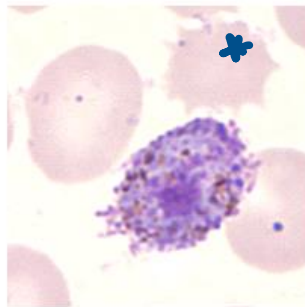
P. ovale gametocytes are round to oval and may almost fill the red blood cells. Pigment is brown and more coarse in comparison to *P. vivax*.



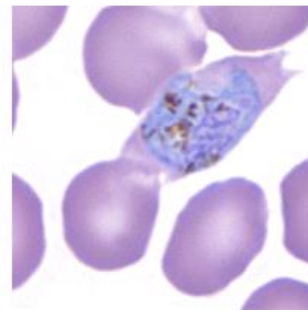
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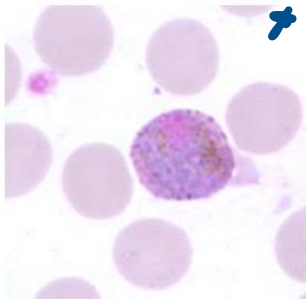
Gametocyte in a thick blood smear.



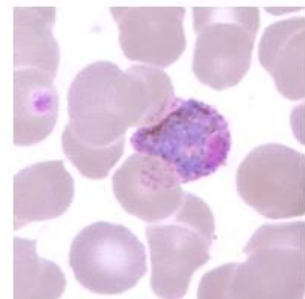
Gametocyte in a thin blood smear.



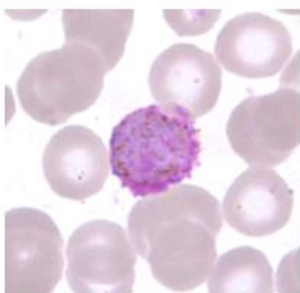
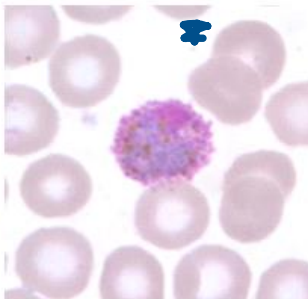
Gametocyte in a thin blood smear. The infected rbc shows some fimbriation.



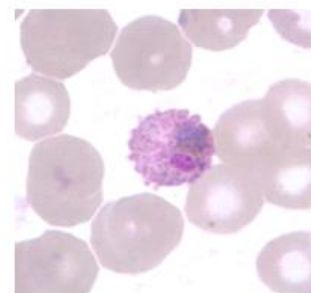
Macrogametocytes in thin blood smears.



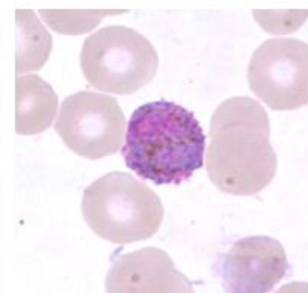
Notice how they nearly fill the infected rbc's. Course pigment, a discrete red nucleus and Schüffner's dots can be seen.



Microgametocyte in thin blood smear. Note the diffuse pigment.



Gametocytes in thin blood smears.



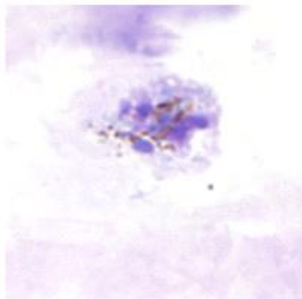
Laboratory diagnosis of malaria

Plasmodium ovale

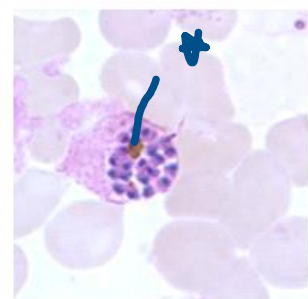
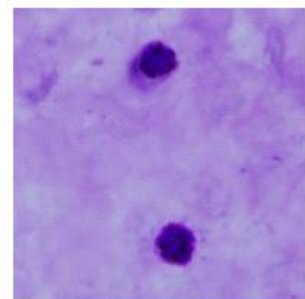
DPD_x
Laboratory Identification of Parasites
of Public Health Concern

4. Schizonts

P. ovale schizonts have 6 to 14 merozoites with large nuclei, clustered around a mass of dark-brown pigment.



Schizonts in thick blood smears.



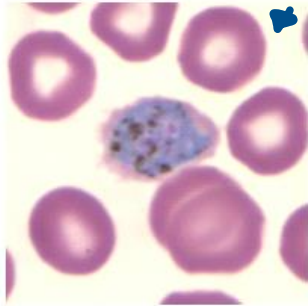
Schizont in a thin blood smear.



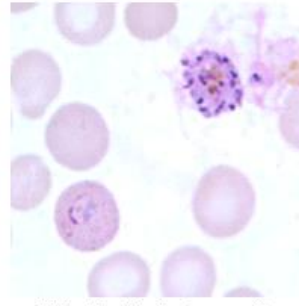
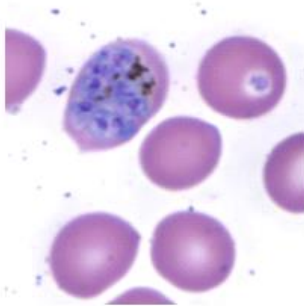
Schizonts in thick blood smears. •



Schizont in a thin blood smear.



Schizonts in thin blood smears. Note the infected rbcs are oval.



Schizont in a thin blood smear with a developing trophozoite.



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Focus

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Plasmodium vivax

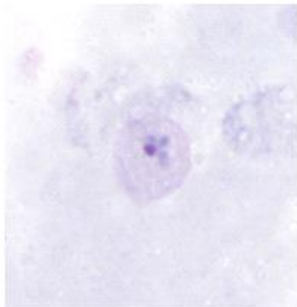
Basic guidelines

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- B. Blood smears, at least two thick and two thin, should be prepared as soon as possible after collection. ***Delay in preparation of smears can result in changes in parasite morphology and staining characteristics.***
- C. Schüffner's dots can be demonstrated in Giemsa stain, which is preferred to Wright or Wright-Giemsa stains.

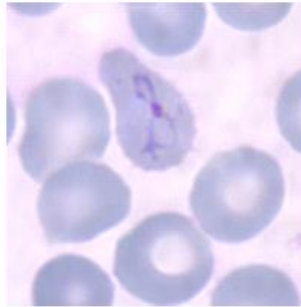
In *P. vivax* infections, red blood cells (rbcs) can be normal to enlarged (up to 1 1/2× to 2×) in size and may be distorted. Under optimal conditions, Schüffner's dots may be seen in Giemsa-stained slides.

1. Rings

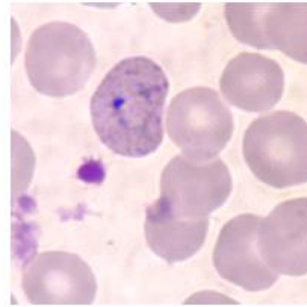
P. vivax rings have large chromatin dots and cytoplasm can become ameboid as they develop.



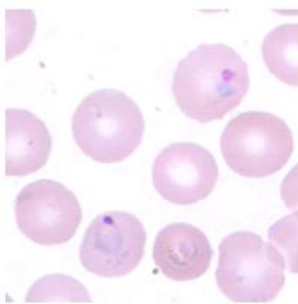
Ring in a thick blood smear. The "halo" is suggestive of Schüffner's dots.



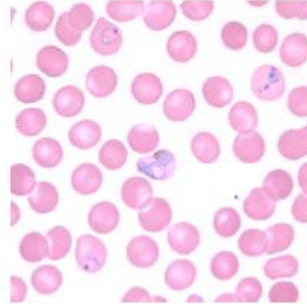
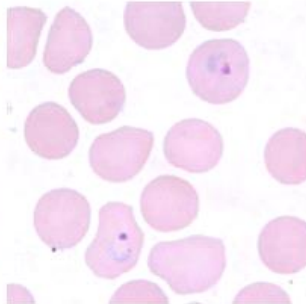
Ameboid ring in an enlarged and distorted infected rbc.



Ameboid ring in an enlarged and distorted infected rbc. Schüffner's dots are visible.



More ring forms in a thin blood smear. Note the slight enlargement of the infected rbcs.

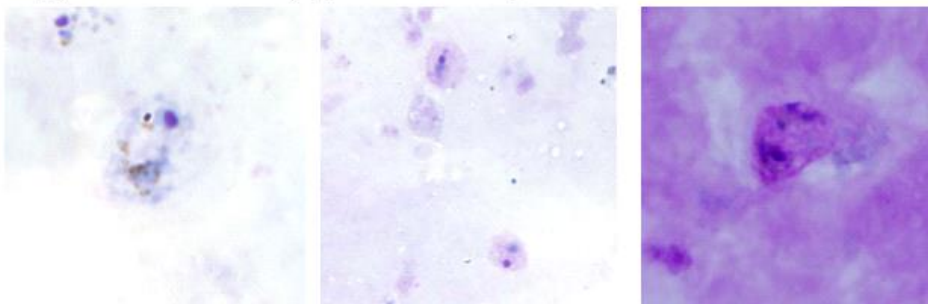


Rings and a trophozoites in a thin blood smear.

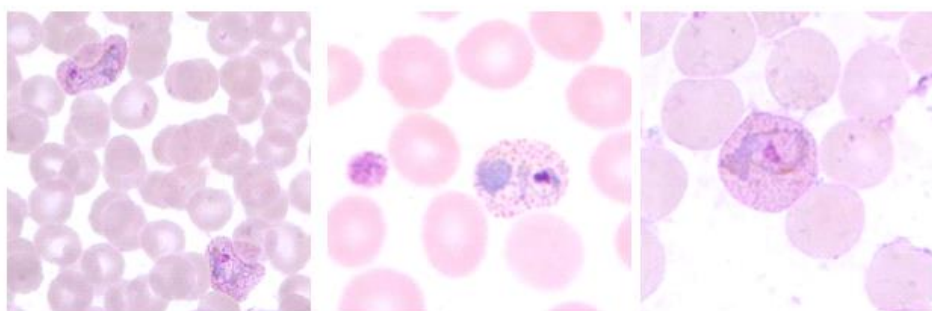
Plasmodium vivax

2. Trophozoites

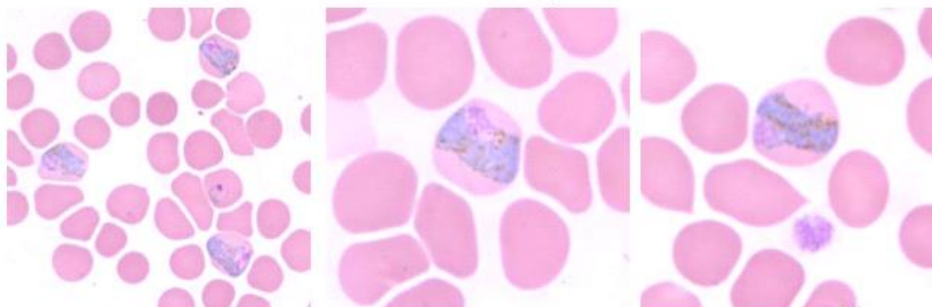
P. vivax trophozoites show amoeboid cytoplasm, large chromatin dots, and have fine, yellowish-brown pigment. Schüffner's dots may appear more fine in comparison to those seen in *P. ovale*.



Trophozoites in thick blood smears.



Large, amoeboid trophozoites in thin blood smears. Note the presence of Schüffner's dots, which are best seen when the blood is stained with Giemsa, and not Wright's stain.

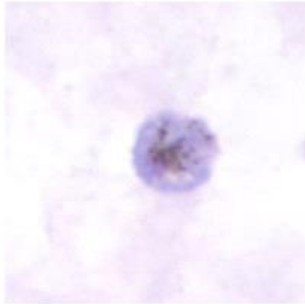


Thin smears showing mature trophozoites. Note the resemblance to the band forms of *P. malariae*. The enlarged size of the infected rbc helps distinguish the two species.

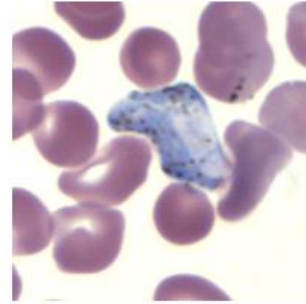
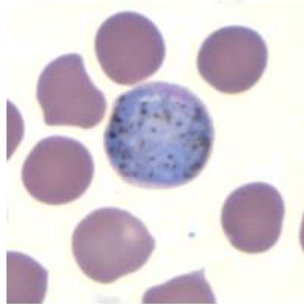
Plasmodium vivax

3. Gametocytes

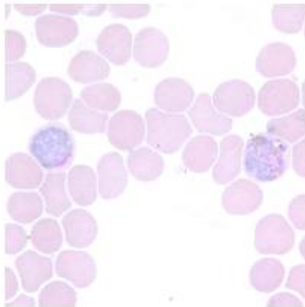
P. vivax gametocytes are round to oval with scattered brown pigment and may almost fill the rbc. Schüffner's dots may appear more fine in comparison to those seen in *P. ovale*.



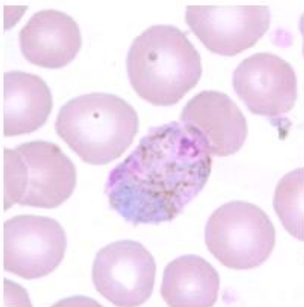
Gametocyte in a thick blood smear.



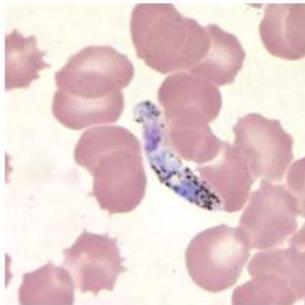
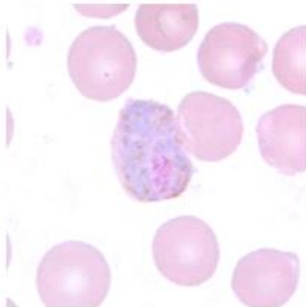
Gametocytes in thin blood smears.



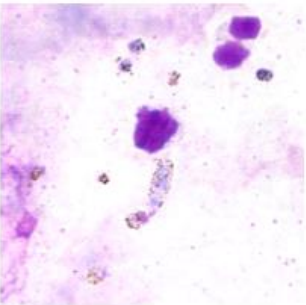
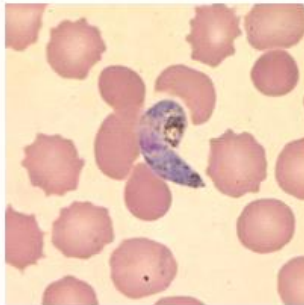
A pair of gametocytes in a thin blood smear.



Gametocytes in thin blood smears. Note the enlargement of the infected rbc and the scattered pigment.



Ookinets in thin blood smears. Ookinets may form if blood is allowed to sit too long before processing.

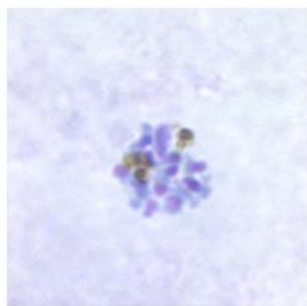


Ookinete in a thick smear.

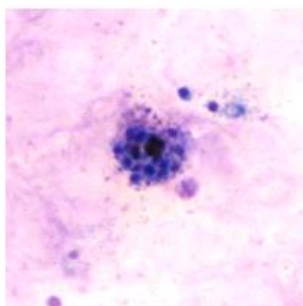
Plasmodium vivax

4. Schizonts

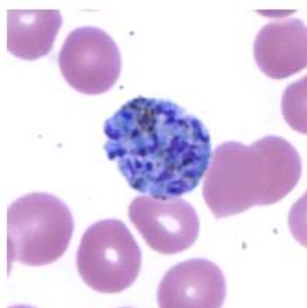
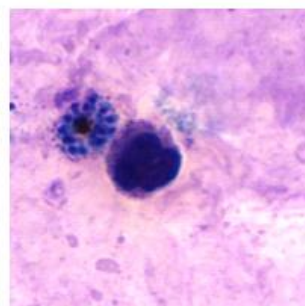
P. vivax schizonts are large, have 12 to 24 merozoites, yellowish-brown, coalesced pigment, and may fill the rbc.



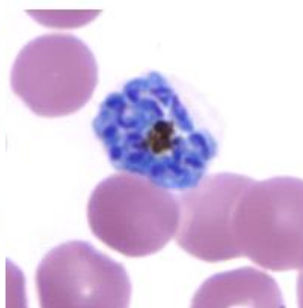
Schizont in a thick blood smear.



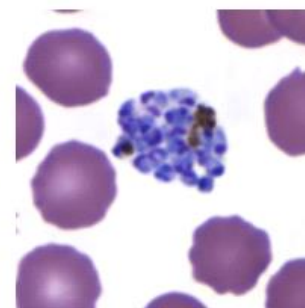
Schizonts in thick blood smears.



Immature schizont in a thin blood smear.



Mature schizonts in thin blood smears.



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Focus

Plasmodium falciparum

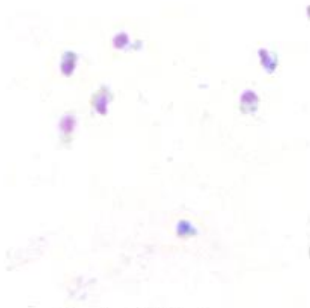
Basic guidelines

- A. Capillary blood should be obtained by fingerstick, or venous blood should be obtained by venipuncture.
- B. Blood smears, at least two thick and two thin, should be prepared as soon as possible after collection. ***Delay in preparation of the smears can result in changes in parasite morphology and staining characteristics.***
- C. Schüffner's dots can be demonstrated in Giemsa stain, which is preferred to Wright or Wright-Giemsa stains.

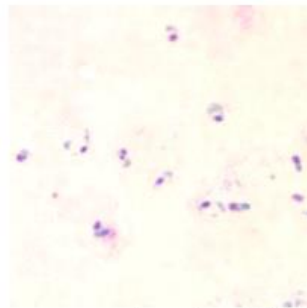
In *P. falciparum* infections, red blood cells (rbcs) are normal in size. Typically only rings and gametocytes are seen unless the blood sat before the smears were prepared.

1. Rings

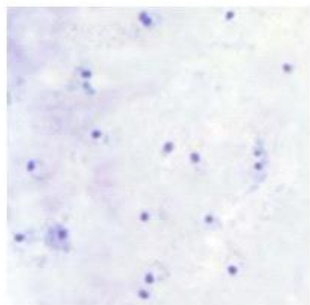
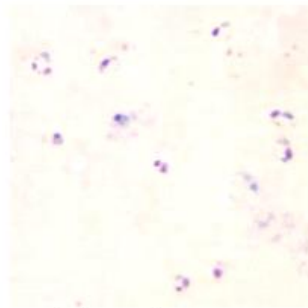
P. falciparum rings have delicate cytoplasm and one or two small chromatin dots. Rbcs that are infected are not enlarged; multiple infection of rbcs is more common in *P. falciparum* than in other species. Occasional appliqué forms (rings appearing on the periphery of the rbc) can be present.



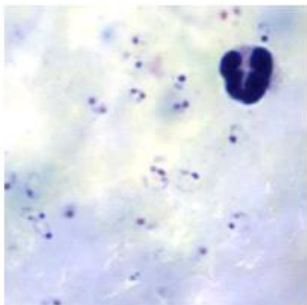
Rings in a thick blood smear.



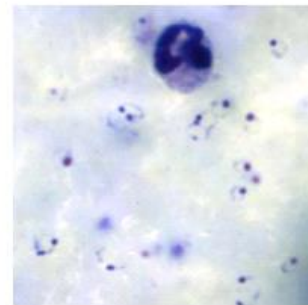
Images from a thick blood smear showing more rings.



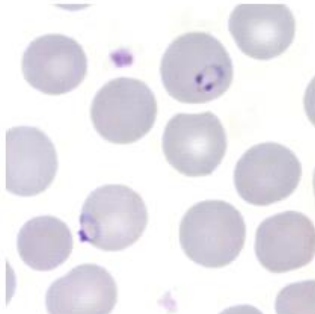
Rings in a thick blood smear.



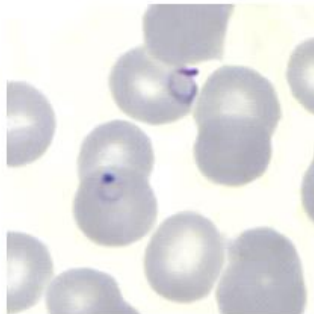
Images from a thick blood smear showing more rings. Note the classic "headphones" appearance of many of the rings.



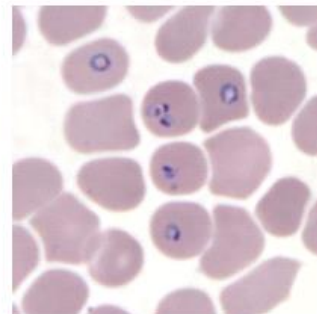
Plasmodium falciparum



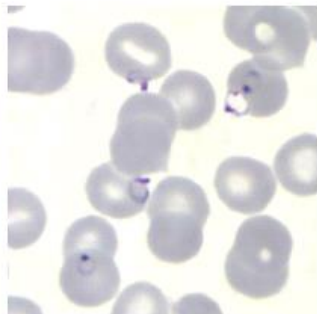
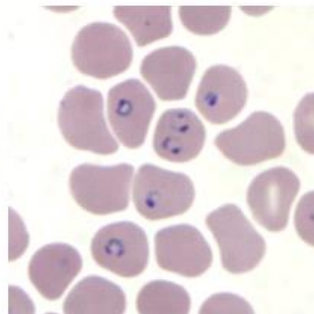
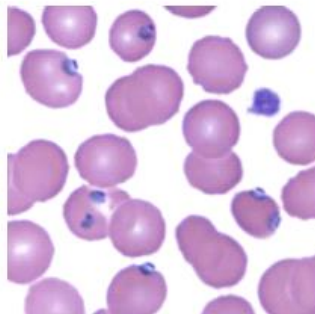
Thin, delicate rings in a thin blood smear. Note the double chromatin dot in the infected rbc at top, and the appliqué form in the infected rbc at bottom.



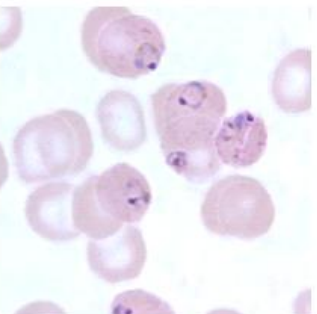
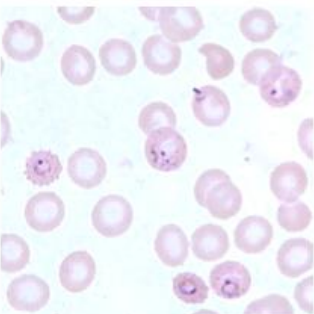
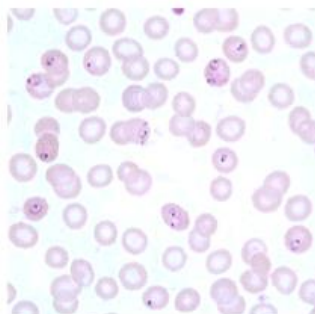
Thin, delicate rings in a thin blood smear.



Rings in a blood smear. Note the multiply-infected rbc's



Rings in thin blood smears. Note the multiply-infected rbc's, some appliqué forms and some classic "head phone" form of several of the infected red blood cells.

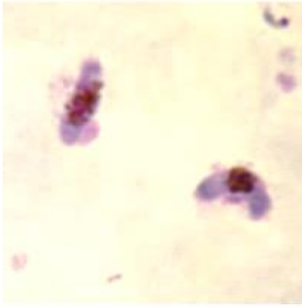


Rings and some developing trophozoites seen in thin smears. Note also the presence of Maurer's clefts, which are often seen in older ring forms. Maurer's clefts stain best with an alkaline pH of 7.2—7.6.

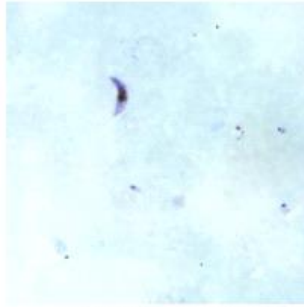
Plasmodium falciparum

2. Gametocytes

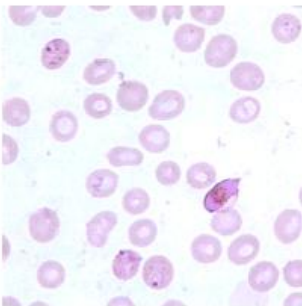
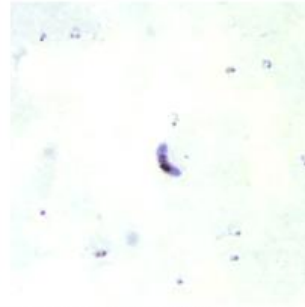
P. falciparum gametocytes are crescent or sausage shaped. The chromatin is in a single mass (macrogamete) or diffuse (microgamete).



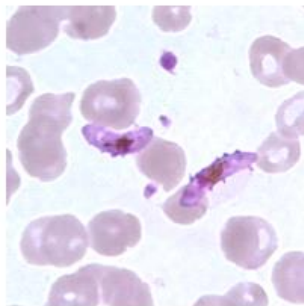
Gametocytes in a thick blood smear.



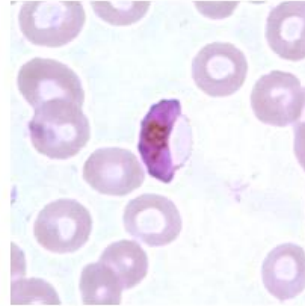
Gametocytes in a thick smear. Note also the presence of several rings.



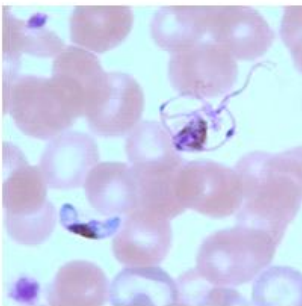
Gametocyte in a thin smear with rings and Maurer's clefts.



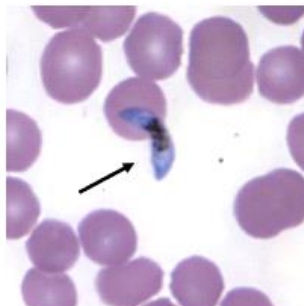
Two gametocytes in a thin smear.



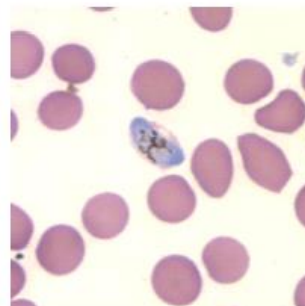
Gametocyte in a thin smear showing the membrane of the rbc.



Gametocytes; the one on the right is undergoing exflagellation

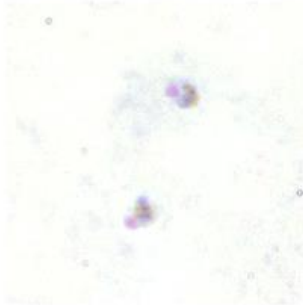


Gametocytes in thin blood smears. Note the presence of "Laveran's bib" (black arrow), which is not always visible.

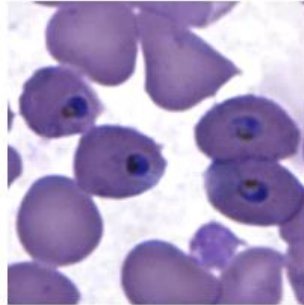


*Plasmodium falciparum***3. Trophozoites**

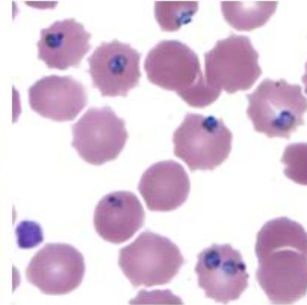
P. falciparum trophozoites are rarely seen in peripheral blood smears. Older, ring stage parasites are referred to as trophozoites. The cytoplasm of mature trophozoites tends to be more dense than in younger rings. As *P. falciparum* trophozoites grow and mature, they tend to retain their ring-like shape and sometimes trace amounts of yellow pigment can be seen within the cytoplasm. Growing trophozoites in *P. falciparum* can appear slightly amoeboid in shape.



Trophozoites in a thick blood smear.



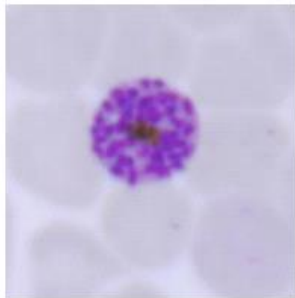
Mature, compact trophozoites in a thin blood smear.



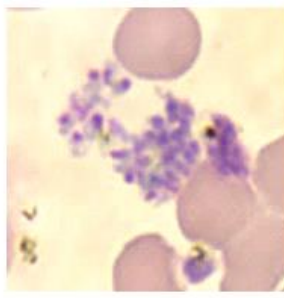
Compact trophozoites in a thin blood smear.

4. Schizonts

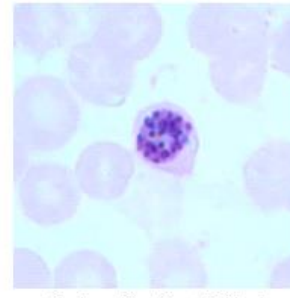
P. falciparum schizonts are seldom seen in peripheral blood. Mature schizonts have 8 to 24 small merozoites; dark pigment, clumped in one mass.



Mature schizont in a thin blood smear.



Ruptured schizont in a thin blood smear.



Another schizont in a thin blood smear.



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Summary

Comparison of the *Plasmodium* Species Which Cause Human Malaria

<i>Plasmodium</i> species	Stages found in blood	Appearance of Erythrocyte (RBC)	Appearance of Parasite
<i>P. falciparum</i>	Ring	normal; multiple infection of RBC more common than in other species	delicate cytoplasm; 1-2 small chromatin dots; occasional appliqué (accollé) forms
	Trophozoite	normal; rarely, Maurer's clefts (under certain staining conditions)	seldom seen in peripheral blood; compact cytoplasm; dark pigment
	Schizont	normal; rarely, Maurer's clefts (under certain staining conditions)	seldom seen in peripheral blood; mature = 8-24 small merozoites; dark pigment, clumped in one mass
	Gametocyte	distorted by parasite	crescent or sausage shape; chromatin in a single mass (macrogametocyte) or diffuse (microgametocyte); dark pigment mass
<i>P. vivax</i>	Ring	normal to 1-1/4 X, round; occasionally fine Schüffner's dots; multiple infection of RBC not uncommon	large cytoplasm with occasional pseudopods; large chromatin dot
	Trophozoite	enlarged 1-1/2-2 X; may be distorted; fine Schüffner's dots	large ameboid cytoplasm; large chromatin; fine, yellowish-brown pigment
	Schizont	enlarged 1-1/2-2 X; may be distorted; fine Schüffner's dots	large, may almost fill RBC; mature = 12-24 merozoites; yellowish-brown, coalesced pigment
	Gametocyte	enlarged 1-1/2-2 X; may be distorted; fine Schüffner's dots	round to oval; compact; may almost fill RBC; chromatin compact, eccentric (macrogametocyte) or diffuse (microgametocyte); scattered brown pigment
<i>P. ovale</i>	Ring	normal to 1-1/4 X, round to oval; occasionally Schüffner's dots; occasionally fimbriated; multiple infection of RBC not uncommon	sturdy cytoplasm; large chromatin
	Trophozoite	normal to 1-1/4 X; round to oval; some fimbriated; Schüffner's dots	compact with large chromatin; dark-brown pigment
	Schizont	normal to 1-1/4 X; round to oval; some fimbriated; Schüffner's dots	mature = 6-14 merozoites with large nuclei, clustered around mass of dark-brown pigment
	Gametocyte	normal to 1-1/4 X; round to oval; some fimbriated; Schüffner's dots	round to oval; compact; may almost fill RBC; chromatin compact, eccentric (macrogametocyte) or more diffuse (microgametocyte); scattered brown pigment
<i>P. malariae</i>	Ring	normal to 3/4 X	sturdy cytoplasm; large chromatin
	Trophozoite	normal to 3/4 X; rarely, Ziemann's stippling (under certain staining conditions)	compact cytoplasm; large chromatin; occasional band forms; coarse, dark-brown pigment
	Schizont	normal to 3/4 X; rarely, Ziemann's stippling (under certain staining conditions)	mature = 6-12 merozoites with large nuclei, clustered around mass of coarse, dark-brown pigment; occasional rosettes
	Gametocyte	normal to 3/4 X; rarely, Ziemann's stippling (under certain staining conditions)	round to oval; compact; may almost fill RBC; chromatin compact, eccentric (macrogametocyte) or more diffuse (microgametocyte); scattered brown pigment

Keypoints for *Plasmodium* Species Which Cause Human Malaria



Infected RBCs

Size	Shape	Schüffner's Dots
<N, N: PM N: PF >N: PO > >N: PV	Crescent: PF (gametocytes) Ameboid: PV Fimbriation: PO Elongated: PO	PV, PO

Parasites Found In Circulating Blood

Rings	Trophozoites	Schizonts (mature)	Gametocytes
Rings only (±gametocytes): PF Numerous: PF Multiply infected RBCs: PF Accessory chromatin dots: PF Delicate: PF	Ameboid: PV Compact: PO PM PF (rarely seen) Band form: PM	6-12 nuclei: PM 6-14 nuclei: PO 12-24: PV 8-24: PF (rarely seen) Rosettes: PM	Crescent: PF Round: PV PO PM

Certain morphologic key characteristics of the infected erythrocytes and parasites can be used to orient the diagnosis towards one of the four *Plasmodium* species that infect humans, as shown above. These characteristics are by no means absolute, however. The final diagnosis should be based on the combined findings for the various characteristics: what is the most probable species, based on the available findings.

Legend

PF: *P. falciparum*
PV: *P. vivax*
PO: *P. ovale*
PM: *P. malariae*